

First Terminal Examination - 2079

Class : XII

Subject: Chemistry (3021)

Full Marks: 75**Time: 3 hrs.****SET 'A'**

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

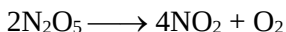
Attempt all the questions.

Group 'A'

Circle the best alternatives.

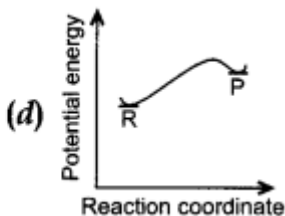
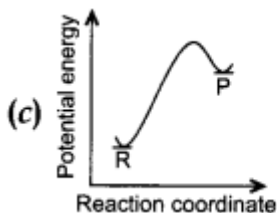
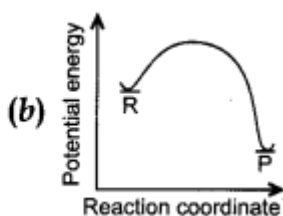
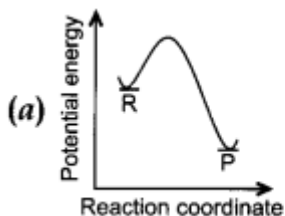
[1 × 11 = 11]

1. What is the concentration of N_2O_5 in the following first order reaction in which the rate is $2.4 \times 10^{-5} \text{ mol/L}$ and rate constant is $3.0 \times 10^{-5} \text{ S}^{-1}$?

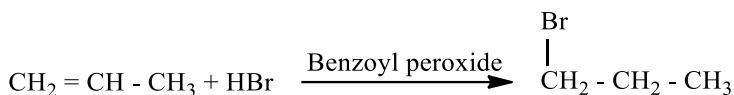


- a. 0.04 b. 0.8
- c. 1.2 d. 1.4
2. If the rate of decomposition of ammonia at 1150 K is found to be $2.10 \times 10^{-6} \text{ mol L}^{-1}\text{s}^{-1}$. The rate of formation of nitrogen is
- a. $1.05 \times 10^{-6} \text{ mol L}^{-1}\text{s}^{-1}$ b. $2.10 \times 10^{-6} \text{ mol L}^{-1}\text{s}^{-1}$
- c. $3.20 \times 10^{-6} \text{ mol L}^{-1}\text{s}^{-1}$ d. None of the above
3. The best method for the conversion of an alcohol into an alkyl chloride is by treating the alcohol with
- a. PCl_5
- b. dry HCl in the presence of anhydrous ZnCl_2
- c. SOCl_2 in presence of pyridine
- d. PCl_3

4. An endothermic reaction with high activation energy for the forward reaction is given by the diagram.



5. C – Cl bond of chlorobenzene in comparison to C – Cl bond in methyl chloride is
- Longer and weaker
 - Shorter and weaker
 - Shorter and stronger
 - Longer and stronger
6. The reaction is an example of



- nucleophilic addition
- free radical addition
- electrophilic addition
- electrophilic substitution

7. The purpose of using anhydrous FeCl_3 as a catalyst in halogenation reaction is to
- initiate the reaction
 - increase the activation energy
 - provide necessary electrophile
 - provide the necessary nucleophile
8. Percentage of carbon in steel?
- 0.1 to 0.25%
 - 2.5 to 5%
 - 0.25 to 2%
 - 2.5 to 4.5%
9. Which of the following is the common name for 2-bromo-2-methylpropane?
- Secondary butyl bromide
 - Neopentyl bromide
 - N-butyl bromide
 - Tertiary butyl bromide
10. Which of the following haloalkane on dehydrohalogenation follows Saytzeff's rule?
- 1-chlorobutane
 - 2-chloropropane
 - 2-chlorobutane
 - 1-chloropropane
11. Which reagent is used to test purity of chloroform?
- AgNO_2
 - NaNO_3
 - AgNO_3
 - Conc. HNO_3

Group - B

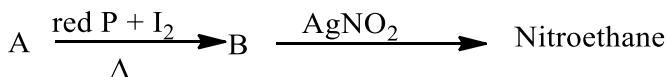
Give short answer to the following questions.

[8 × 5 = 40]

1.
 - a. Define average rate of reaction. [1]
 - b. The decomposition of ammonia on tungsten surface takes place as
$$2\text{NH}_3(\text{g}) \rightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$$

Write the rate expression for each component. [1]
 - c. Derive integrated rate equation for first order reaction. [3]
2.
 - a. The rate of reaction doubles when concentration of "P" doubles where as there is no effect on the rate of reaction with the change in concentration of "Q". What is the order of reaction with respect to P and Q and overall order? Write rate law for the reaction. [2]
 - b. Define first order reaction with an example. In a first order reaction, 20% of the reactant is consumed in 25 minutes. What is the half life period? [1+2]
3.
 - a. How does catalyst affect the rate of chemical reaction? Illustrate with the help of energy profile diagram. [2]
 - b. If the temperature coefficient of a reaction is 2. What will be the rate of reaction when temperature is increased from 30°C to 100°C? [1]
 - c. Define the term effective collision. Mention any two conditions for such collision. [2]
4. Zinc is most commonly used to galvanize other metals.
 - a. Draw a well labelled diagram of vertical retort with reaction involved in the (extraction) reduction of zinc [2]
 - b. How does zinc sulphate reacts with excess caustic soda solution? [2]
 - c. How is lithopone prepared from white vitriol? Write its use. [1]

5. A metal forms two different chlorides 'X' and 'Y'. 'X' gives black precipitate with ammonia, while 'Y' gives white precipitate. With potassium iodide, 'Y' gives scarlet (red) precipitate which is soluble in excess of potassium iodide.
- Name the compound 'X' and 'Y' [1]
 - Give any one preparation method for both compound 'X' and 'Y' [2]
 - State the chemical equation of compound 'X' with ammonia. [1]
 - Give any one reaction of compound 'Y' acting as an oxidizing agent. [1]
6. a. What is Williamson's synthesis? Give one example. Also write its demerit. [3]
- b. Starting from trichloromethane how would you prepare;
i) chloropicrin ii) Methane [2]
7. What is Sandmeyer's reaction? Give an example. Why does chlorobenzene give o and p-substituted products on electrophilic substitution though it is a deactivating group? Convert benzene to benzoic acid? [1+2+2]
8. a. How chloroform is prepared from propanone and bleaching powder paste? [1.5]
- b. Why is chloroform stored in dark colored bottles containing ethanol? [1.5]
- c. Identify A and B in the following reaction sequence with their IUPAC names. [2]

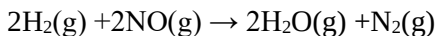


Group - C

Give long answer to the following questions.

[3 × 8 = 24]

9. Hydrogen and nitrogen oxide react according to the following equation:



Experiments were performed at 800°C in order to determine the order of reaction and the following results were obtained.

Initial concentration of nitrogen oxide (mol litre ⁻¹)	Initial concentration of hydrogen (mole litre ⁻¹)	Initial rate of production of nitrogen (mol litre ⁻¹ sec ⁻¹)
6×10^{-3}	1×10^{-3}	3×10^{-3}
6×10^{-3}	2×10^{-3}	6×10^{-3}
1×10^{-3}	6×10^{-3}	0.5×10^{-3}
2×10^{-3}	6×10^{-3}	2×10^{-3}

- Differentiate order from molecularity. [2]
 - What is the order of this reaction with respect to
(i) nitric oxide (ii) hydrogen [2]
 - Write the rate equation for the reaction of nitric oxide with hydrogen. [1]
 - What is the value and unit of rate constant, K? [1]
 - What is the rate of reaction if $[\text{H}_2] = 0.5\text{M}$ and $[\text{NO}] = 0.1\text{M}$ [1]
 - Why higher order of reaction not so common? [1]
10. In the extraction of copper from copper pyrites, the ore is first concentrated by froth floatation, and then heated in a furnace.
- How is blister copper obtained from copper matte? [3]
 - Write the action of heat on blue vitriol. [2]
 - How does blue vitriol react with excess sodium hydroxide solution? [2]
 - Write the formula and one use each of black and red oxide of copper [1]

11. a. A primary iodoalkane 'A' reacts with NaCN to produce compound 'B'. 'B' on hydrolysis with dil. HCl produces compound 'C'. Compound 'A' on heating with Na and ether produces butane. Identify A, B, C and write reactions involved. [3]
- b. Why do alkyl halides show nucleophilic substitution reactions as their major reactions? What are SN^1 and SN^2 reactions? Why iodomethane prefer SN^2 mechanism? [1+3+ 1]

ॐ - The End - ॐ